

Teorema de Noether

$$L(q, \dot{q}) = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} k q^2$$

$$t \mapsto q(t)$$

$$\left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \cancel{\frac{\partial L}{\partial \lambda}} = 0$$

0 cuando No hay ligadura

$$\perp \quad -kq - m\ddot{q} = 0$$

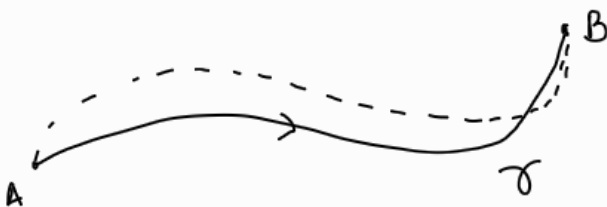
Principio de Mínima Acción

$$q = (q_1, q_2, \dots, q_N) \in \mathbb{R}^N$$

$$L: \mathbb{R}^N \times \mathbb{R}^N \longrightarrow \mathbb{R}$$

$$(q, \dot{q}) \longmapsto L(q, \dot{q})$$

Supongamos escoger una trayectoria



$$t \longmapsto \gamma(t) = (x_1(t), x_2(t), \dots, x_n(t))$$

Velocidad $\dot{\gamma}(t) = (\dot{x}_1(t), \dots, \dot{x}_n(t))$

Cond: $\gamma(t_i) = A$
 frontera $\gamma(t_f) = B$

Acción:

$$S[\gamma] = \int_{t_i}^{t_f} L(\gamma(t), \dot{\gamma}(t)) dt$$

Variaciones de la acción " $\delta S = 0$ "

